claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a639ecbd59c4115b8.jsonl`
- SHA-256: `7610d350c733246511b84a8b2ea94f848b11c3ae26f3a2fd37f20ea79f4c62d1`
- Source modified: `2026-06-06T17:31:09+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:30:34.659Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://arxiv.org/html/2603.13397v2
Title: TennisExpert: Towards Expert-Level Analytical Sports Video Understanding
Found via: recent/applied search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:37.654Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-06T17:30:51.933Z)

TennisExpert: Rally Segmentation Analysis

(1) Temporal Segmentation Methods

The paper employs **audio-based hit detection** for rally segmentation:

> "we train a lightweight audio neural network to identify the distinctive acoustic signatures of racquet?ball impacts"

The process involves: detecting racquet-ball impact signatures from broadcast audio, clustering these temporal anchors to form candidate rally intervals, then filtering for standard broadcast views and visible scoreboards.

(2) Quantitative Segmentation Accuracy

The paper provides **no F1 or accuracy metrics for rally segmentation itself**. It only reports segmentation completion: "segmented into 40,523 rally-level clips" from 202 matches (471.9 hours). Event spotting accuracy is mentioned separately: "the detector...achieving an Edit Score of 81.2 on the test set" (for fine-grained event detection within rallies, not rally boundaries).

(3) Cloud vs. Local Models & Compute

The framework uses **local fine-tuned models** with modest compute:
- Qwen3-VL-8B backbone (8 billion parameters)
- Inference: "~20 GB VRAM" and "< 2 s latency"
- Training: "4 NVIDIA H200 GPUs"

No cloud VLM dependency for inference, though dataset construction used "Gemini 3 Pro" for commentary synthesis.

(4) Datasets & Licenses

Only TennisVL (their own) is used for training/evaluation. No mention of ShuttleSet, TrackNet, or badminton datasets. The paper states: "dataset is compiled entirely from publicly accessible broadcast videos sourced from YouTube" under "CC BY 4.0" license with usage restricted to "academic research."

(5) Publication Date

arXiv:2603.13397v2 [cs.CV] 17 Mar 2026

(6) Teacher-Student Distillation

No mention of teacher-student distillation anywhere in the paper.

4. assistant (2026-06-06T17:30:54.970Z)

This source is directly relevant on multiple dimensions of the research question: audio-based hit detection for rally segmentation, local-model viability, and the use of a cloud VLM (Gemini 3 Pro) for dataset construction rather than runtime inference. Let me record the structured findings.

5. user (2026-06-06T17:31:04.156Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:09.260Z)

Extracted 5 falsifiable claims from the TennisExpert paper (arXiv:2603.13397v2, primary research, published 2026-03-17). The most relevant finding: it uses a lightweight audio neural network detecting racquet-ball impact signatures as the core rally-segmentation primitive (central to the research question's audio-based hit-detection thread), and it confirms the cloud-VLM-offline / local-model-runtime split (Gemini 3 Pro for dataset construction, local Qwen3-VL-8B for inference). Caveats noted: no rally-boundary F1 reported, ~20 GB VRAM exceeds the RTX 2070 Super's 8 GB, and it is tennis-only with no badminton/ShuttleSet/TrackNet usage and an academic-only CC BY 4.0 dataset.